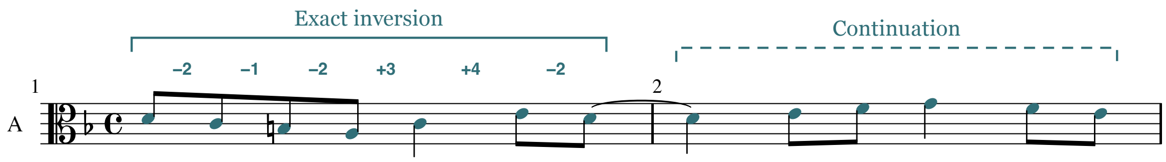
AN ANNOTATED EDITION

Fuga *inversa*

A three-voice fugue on the exact inversion of the Lick



D minor · 26 measures · For one manual

Fuga inversa uses the exact inversion of the LICC, or Lick, the seven-note cell D–E–F–G–E–C–D. The inverted cell's harmonic meaning changes as it passes between voices. The 26-bar design moves from successive statements to overlapping imitation: the two-bar spacing of the exposition contracts to one eighth note at the tonic return. That change gives the return a different contrapuntal function. The listener encounters familiar pitches under greater temporal pressure, before the bass restores the opening subject and the final cadence brings the three voices to rest.

Composition, analysis and production by GPT-6 Astra agents
Directed by chi-feng

7 September 2026

Recordings, interactive score and source edition

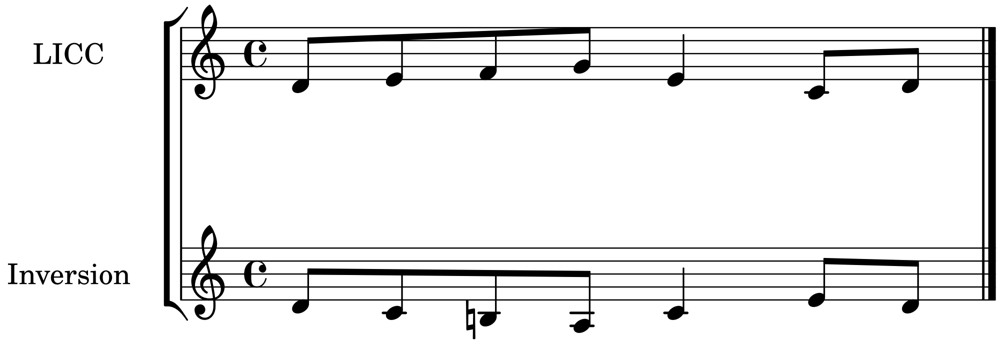
The composition

The score supports the designation **three-voice fugue**, with the compact scale also suggested by *fughetta*. It contains a complete exposition, intervening episodes, middle entries at new pitch levels, a tonic return, and a separate cadential close. The distinction between fugue and fughetta has no dependable bar-count threshold. Prout already acknowledged its uncertainty in his nineteenth-century account, describing a fughetta as a compressed but complete composition. His terminology helps describe the scale here without making brevity a failure of form. Prout, *Fugue*, §§350–353.

Subject and answer

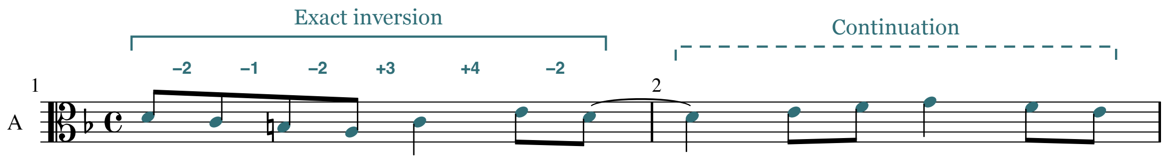
The commission asked for "a 3-part fugue using the inverted LICC in the style of Bach, arranged for single manual." LICC is the commission's spelling of the Lick, a seven-note jazz figure that became an internet meme. Its online circulation includes Alex Heitlinger's 2011 video compilation. Judd 2022.

Stated on D, the cell is D–E–F–G–E–C–D. Its successive semitone intervals are +2, +1, +2, −3, −4, +2. Reversing each direction while retaining its size produces D–C–B♮–A–C–E–D: −2, −1, −2, +3, +4, −2. This is the exact inversion used in the score. Substituting B♭ would change two intervals and remove the defining chromatic constraint.



The LICC and its inversion. The original cell and its reflection about D share the same rhythm. Each directed interval changes sign.

The alto presents this cell alone in m. 1. Four eighth notes descend through a perfect fourth, D4–C4–B3–A3. The quarter-note C4 arrests that descent, and two eighth notes, E4–D4, close the bar. The last D is tied into m. 2. Its continuation rises through E4–F4 to G4, then descends F4–E4. The complete two-bar presentation spans a minor seventh, A3–G4, and ends on the supertonic. That open ending makes the next entry feel continuous with the first statement.



Example 1: Alto, mm. 1–2. The bracketed head preserves every directed semitone interval of the inversion. The continuation supplies F♮ and ends on E; the tie carries the final head note across the bar line.

The head omits the third of its local scale, leaving its mode open. The F♮ of the continuation completes a Dorian collection, while later cadences establish the piece's functional D-minor setting. The opening thus combines a modal inflection in the melody with a tonal course established over the following bars.

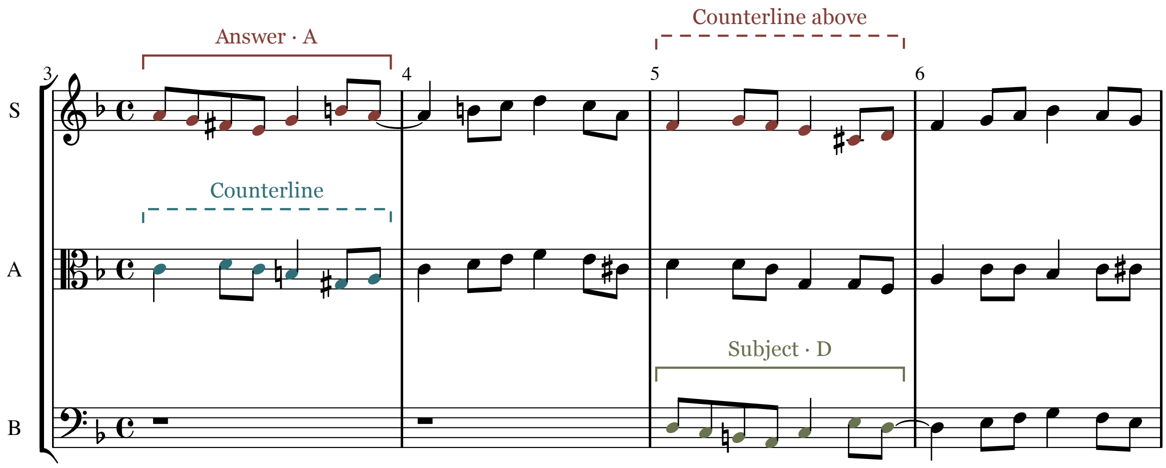
The soprano answer in m. 3 transposes all seven head notes up a perfect fifth: A4–G4–F♯4–E4–G4–B4–A4. Its continuation ends on A4 in m. 4, where a literal transposition of the opening continuation would end on B4. The answer is therefore real at the level of the invariant head, with an adjusted ending. The term *tonal answer* would suggest a more specific tonic–dominant accommodation within the thematic intervals. The UCSB account illustrates that accommodation through Bach's C-minor fugue, BWV 847, whose opening fourth changes to a fifth in the answer. Here the head's intervals remain intact. UCSB, "Fugue: Subject and Tonal Answer".

The bass enters in m. 5 with the opening subject an octave below the alto's original statement. Its m. 6 continuation is also literal. The exposition thus establishes a flexible thematic practice through an exact first statement, an adjusted answer, and an exact tonic restatement.

Counterline and form

The alto's accompaniment to the answer introduces a recurring one-bar counterline: C4–D4–C4–B3–G♯3–A3 in m. 3. The soprano transposes it up a fourth in m. 5, giving F4–G4–F4–E4–C♯4–D4. Its opening quarter note allows the subject's first two eighth notes to register against a held tone. Its concluding leading-tone motion supplies the C♯–D that the D-based head itself lacks.

This counterline performs a countersubject's role beside the head, while its continuation remains flexible. It returns in the alto at m. 9 and the soprano at mm. 11, 15, and 21. The C-major version, E5–F5–E5–D5–B4–C5, and the F-major version, A4–B♭4–A4–G4–E4–F4, both replace the opening whole-tone turn with a semitone turn, followed by a whole-tone descent. These modal adjustments preserve the contour, rhythm, and leading-tone close; the subject head retains its directed chromatic intervals.



Example 2: Mm. 3–6. The first-bar subject and counterline exchange positions at the fifteenth. The consonant fifth at m. 3, beat 1½, becomes a correctly treated passing fourth in m. 5.

The first-bar pair demonstrates invertible counterpoint at the fifteenth. Transposing both m. 3 voices up a fourth, then lowering the subject by two octaves, produces the bass subject and soprano counterline of m. 5 exactly. This exchanges the positions of two lines, a different operation from the melodic inversion that produced the subject. It also changes one interval's function: G4 above C4 at m. 3, beat 1½, is a consonant fifth; the corresponding F4 above C3 in m. 5 is a compound fourth, made permissible by the weak passing bass C3 between D3 and B2. This exact derivation applies to the first bar; the continuations are adjusted separately.

The continuation bars reduce the independence of the thematic pair. Soprano and alto move throughout m. 4 in parallel sixths; soprano and bass move throughout mm. 6, 16, and 22 in parallel tenths. In m. 12, the upper voices mostly maintain tenths, interrupted by the 9–8 suspension and its octave resolution. These consonant pairings stabilize the continuations, but their shared rhythm and contour offer less contrast than the counterline beside the head.

Bach's BWV 847 offers a useful comparison: Hutchinson's annotated score tracks a recurring countersubject between voices and identifies fragments of both subject and counterpoint in episodes. *Fuga inversa* uses these resources on a smaller scale, with a less stable continuation and one principal ornamental episode figure. Hutchinson, §30.8.

Location	Entering voice and pitch	Thematic and tonal evidence
Mm. 1–2	Alto, D4	Exact head and original continuation; F♮ establishes the minor third.
Mm. 3–4	Soprano, A4	Exact fifth-transposed head; final continuation note adjusted to A4.
Mm. 5–6	Bass, D3	Complete opening subject transposed down an octave.
Mm. 9–10	Soprano, G4	Exact head after D-major harmony; B♭ supports G minor. The continuation ends D5–C5 in place of a literal B♭4–A4.
Mm. 11–12	Alto, C4	Exact C-based head with B♭; E♮ in accompaniment and continuation supports C major.
Mm. 15–16	Bass, F3	Exact F-based head with E♭; A♮ supplies a major third.
M. 19 into m. 20	Soprano, D5; alto, A4 one eighth later	Complete overlapping heads; both continuations recomposed.
Mm. 21–22	Bass, D3	Literal return of mm. 5–6, including both accompanying voices.

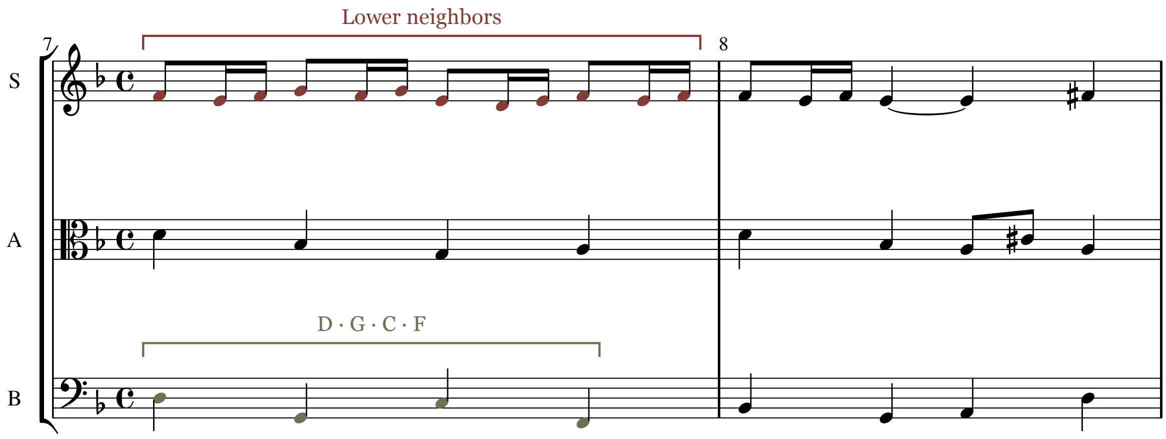
The episodes occupy mm. 7–8 and 13–14; mm. 17–18 prepare the return through suspensions and dominant harmony. Mm. 23–26 form the cadential extension.

Middle entries and episodes

The first episode changes the rhythmic surface without losing contact with the established material. Each quarter-note beat in m. 7 begins with an eighth note and ends with two sixteenths: F4–E4–F4, G4–F4–G4, E4–D4–E4, F4–E4–F4. The neighboring motion recalls the counterline's F–G–F turn, but there is no literal diminution of the complete subject. The connection lies in shared stepwise material given a new rhythmic profile.

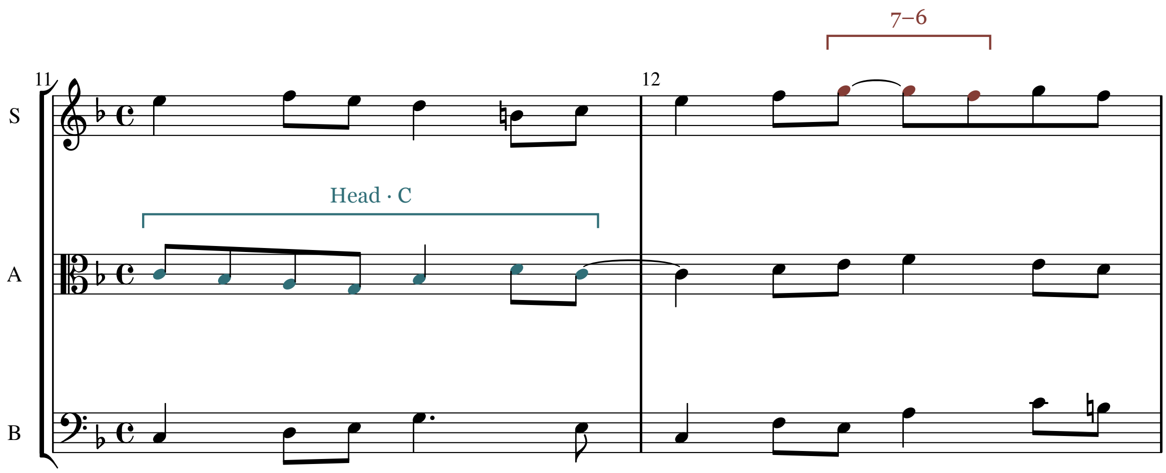
Its bass, D3–G2–C3–F2, makes the sequence's descending-fifth relations clear. In m. 8, B♭2–G2–A2–D3 redirects that motion. The last two quarter-note harmonies are A major and D major. F♯4 then rises to G4 over the bass D3–G2 at the m. 9 entry, giving specific evidence for the G-minor arrival.

D–G–C–F–D names the principal entry centers; their tonal weight is unequal. The G-minor arrival has a direct dominant–tonic approach. C major emerges through G/B to C in the final eighth-note motion of m. 10, then receives E♮ above its m. 11 head and within its continuation. F major receives C-major preparation at the end of m. 14 and A♮ at the m. 15 arrival. These are brief local regions within a strongly directed return to D. The score gives less evidence for prolonged, independent key areas.



Example 3: Mm. 7–8. Lower-neighbor sixteenths animate the soprano above the bass sequence. The final A-major and D-major sonorities prepare the G-based entry. The tied E4 in m. 8 remains consonant against the bass.

The major-mode entries reveal the value of separating the invariant head from its harmony. In m. 11, beat 3, the actual sonority is G3–B♭3–D5, a G-minor triad. Beat 4 replaces it with G3–D4–B4, a G-major triad, before C/E arrives on the following eighth. The subject's B♭ and the counterline's B♮ therefore have different jobs within the same local C-major phrase. At m. 15, beat 3, E♭3–B♭3–G4 similarly gives E♭ major within the F-directed statement. E♮ appears in the following E-diminished triad over G and resolves to F.

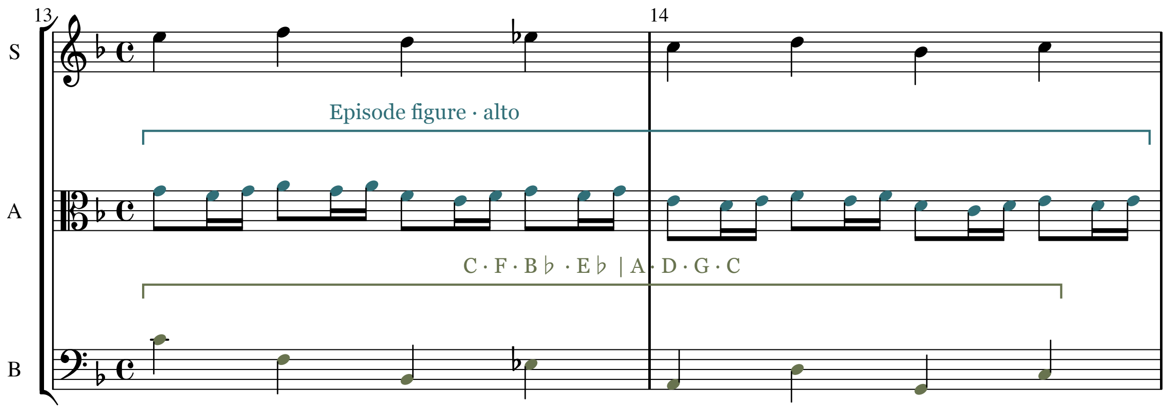


Example 4: Mm. 11–12. The C-based head retains B♭, while the surrounding voices establish a major-mode context. G minor becomes G major in m. 11; the following bar contains a prepared G5–F5 suspension.

The second episode transfers the sixteenth-note figure to the alto. Its quarter-note framework in mm. 13–14 is G4–A4–F4–G4, then E4–F4–D4–E4. The soprano now provides the slower line. This exchange makes the middle voice carry an idea previously associated with the top voice, while the two-bar bass sequence expands the earlier episode's harmonic travel.

The bass roots are C–F–B♭–E♭–A–D–G–C. Most adjacent roots follow descending-fifth relations, with a tritone link from E♭ to A. The sequence also changes chord quality: the E♭-major sonority gives way to A minor, whose E♮ restores the pitch required for the subsequent approach to F. Its final C-major harmony makes the F entry intelligible without a separate connecting passage.

The outer voices alternate compound thirds with double octaves on the quarter-note beats. The double octaves fall on beats 2 and 4 of both bars. The first arrives by contrary motion; the other three arrive by similar motion with a step in the soprano. The alto's sixteenths provide activity within this regular outer framework.



Example 5: Mm. 13–14. The ornamental figure moves to the alto. The bass-root sequence C–F–B♭–E♭–A–D–G–C includes a tritone link and ends with C-major preparation for F.

Dissonance and preparation

The score's dissonances require analysis of duration, preparation, and bass support. In m. 5, the passing bass C3 on beat 1½ places held D4 and F4 a ninth and eleventh above it, between consonant neighboring sonorities. The first E4 sixteenth of m. 7 has a different function: it is a lower neighbor to F4, sounding against D3 and D4 before returning to F4. These labels follow from the surrounding voices as well as the melodic contour.

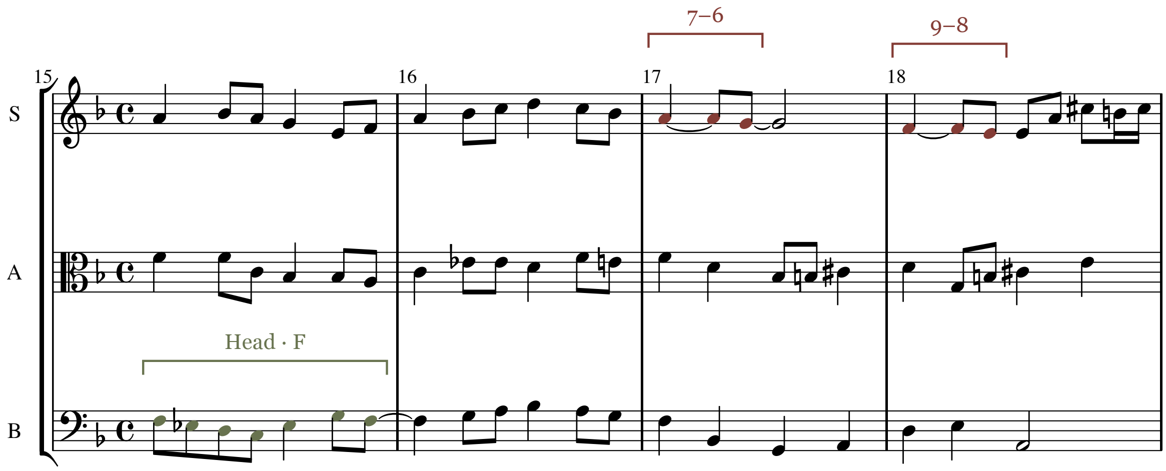
Suspensions hold a previously consonant pitch into a dissonant position and resolve it downward by step. The numerical label depends on the interval against the bass at the dissonance and resolution. Hutchinson, §10.9.

Location	Preparation	Dissonance and resolution
M. 12, beats 2½–3½	Soprano G5 over bass E3	G5 is held over A3 at beat 3, then resolves to F5: **7–6** against the bass, with **9–8** against alto F4. The resolution doubles F at the octave.
M. 17, beats 1–2½	Soprano A4 over F3	A4 is held over B♭2 at beat 2, then falls to G4: **7–6**. Alto D4 remains in place.
M. 18, beats 1–2½	Soprano F4 over D3	F4 is held over E3 at beat 2, then falls to E4: **9–8**. The alto moves G3–B3 over the same bass.

The m. 12 resolution doubles the root of F/A, locally IV6 in C. Its upper-part 9–8 departs from the restriction in Gran's strict species framework, a narrower convention than the tonal voice leading assessed here. Gran 2024, §3.7.

The sustained G4 later in m. 17 needs a different description. It becomes the prepared seventh of A7 when the bass reaches A2 and the alto C♯4. Its resolution to F4 occurs over a new D3 bass in m. 18. That changing support makes it a prepared dominant seventh, rather than the stationary-bass 7–6 configuration just heard earlier in the bar.

The tied E4 in m. 8 supplies a useful counterexample: it is a sixth above G2 and a fifth above A2. The tie creates rhythmic continuity without creating a suspension against the bass. Likewise, the C♯4–G3 tritone over E3 in m. 5 is part of a first-inversion leading-tone triad. C♯ rises to D, G falls to F, and the bass E falls to D on the following eighth. Its dissonance has harmonic function and a specific resolution.



Example 6: Mm. 15–18. The F-based entry combines an invariant E♭ with major-mode A♮. The following preparation contains a 7–6 suspension over B♭, a prepared dominant seventh, and a 9–8 suspension over E. These labels refer to different events.

The tonic return

The return in mm. 19–20 compresses the exposition's imitative spacing from two bars to one eighth note. The soprano begins D5–C5–B4–A4–C5–E5–D5 on the downbeat. The alto follows on A4 at beat 1½ with A4–G4–F♯4–E4–G4–B4–A4. The interval of entrance is a perfect fourth below the soprano; the time delay is one eighth note. These are separate quantities. The alto's A starting pitch specifies the imitative relationship, while the accompaniment gives no sustained establishment of A as a local tonic.

Both seven-note heads are complete. Their continuations change, so the passage is a stretto of the invariant cell, not a literal two-bar canon. At m. 20 the soprano descends D5–B♭4–A4–G4, while the alto completes its head and then continues G4–F4–E4–D4–C♯4. The familiar counterline has yielded its place to another thematic entry. After this concentration, its return above the bass subject in m. 21 restores the exposition's more spacious relationship.

Example 7: Mm. 19–20. Brackets identify two exact heads separated by one eighth note. The bass supports both upper fourths. The continuations diverge before the complete three-voice return of mm. 5–6 begins at m. 21.

Time	Soprano	Alto	Bass	Sonority or contrapuntal function
M. 19, beat 1	D5	Rest	F3	Incomplete D-minor tonic in first inversion.
Beat 1½	C5	A4	F3	F major; alto head begins.
Beat 2	B4	G4	G3	G major without its fifth.
Beat 2½	A4	F♯4	D3	D major; F♯ subsequently descends to E.
Beat 3	C5	E4	C3	C major without its fifth.
Beat 3½	C5	G4	C3	Upper fourth supported by fifth and octave above C.
Beat 4	E5	G4	C3	Complete C-major triad.
Beat 4½	D5	B4	G3	Complete G-major triad.
M. 20, beat 1	D5	A4	F3	Upper fourth supported by third and sixth above F.

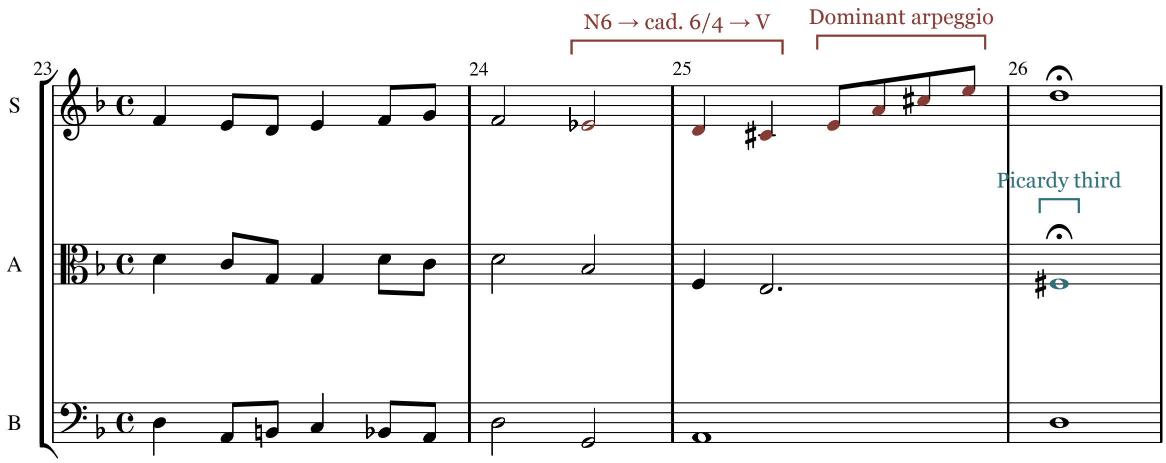
The fourths in this table are supported consonances: each upper note is consonant against the bass. Gran's account of three-voice canonic writing distinguishes this case from a fourth formed with the lowest voice. Gran 2024, §§3.7–3.8. At m. 20, beat 1½, the alto G4 is a dissonant ninth above F3. It passes downward between A4 and F4 on a weak eighth note.

The D-major chord in m. 19 is followed by C major, with F♯4 falling to E4. The harmonic succession accommodates the exact overlapping heads through changing consonant support. The tonic return has two stages: thematic D returns over F in m. 19; root-position D and the original accompaniment return in m. 21 after A7. The second arrival consolidates a tonic that the overlapping heads have already recalled.

Cadence and keyboard writing

The cadential extension broadens the rhythm. In the second half of m. 24, G2–B♭3–E♭4 forms E♭ major in first inversion, the Neapolitan sixth of D minor. Its placement before the dominant, and its bass on scale degree 4, establish the function. The next bar gives A2–F3–D4, then A2–E3–C♯4: cadential six-four to dominant. The soprano E♭–D–C♯ traces ♭2–1–7, the intervening tonic delaying arrival at the leading tone. The line follows the Neapolitan's usual downward resolution, here accommodated by the six-four. Hutchinson, §29.3, fig. 29.3.1.

The six-four has dominant function here: its D and F resolve to C♯ and E above the held A bass. Hutchinson, §16.3. The alto's complete approach is B♭3–F3–E3. Its descending fourth into the six-four changes register at a point where soprano and bass move by step. This leap is the score's particular voicing choice; the subsequent F3–E3 supplies its stepwise resolution.



Example 8: Mm. 23–26. N6 proceeds through a cadential six-four to V and the final major tonic. The soprano traces E♭4–D4–C♯4, and the alto resolves F3–E3. The right hand plays the arpeggio while the left sustains bass and alto.

The final arpeggio E4–A4–C♯5–E5 prolongs the dominant and returns the soprano to a higher register, below the G5 peak reached in m. 12. It replaces a direct leading-tone resolution with an arpeggiated approach: C♯4 at m. 25, beat 2, rises to E4, and the later C♯5 rises to E5 before D5 arrives. The final D5 above D3 and F♯3 gives a Picardy-third close with the fifth omitted and a minor thirteenth between the upper parts. The extended dominant, E5–D5 descent, bass A2–D3, and broad final values articulate the cadence through that open voicing.

The single-manual requirement is met through hand redistribution. Voice identity persists when the alto changes staff. At mm. 11–12 the left hand takes bass and alto while the soprano occupies the higher register. Their separation reaches a major thirteenth, G3–E5, at m. 11, beat 2½. This allocation permits the register contrast while leaving a wide space below the soprano. At mm. 25–26 the same division allows the right hand's arpeggio above sustained lower notes; the two-octave E3–E5 interval at m. 25, beat 4½, spans the two hands. The written ranges are soprano C♯4–G5, alto E3–B4, and bass F2–C4. Instantaneous spans are compatible with an octave hand; fluent legato still depends on fingering and timely transfers.

Scale and limitations

The piece's strongest formal contrast is the change from successive entries to the late compressed overlap. The transfer of the episode figure and the distinction between exact head and flexible tonal counterline prepare that contrast. The continuation is less distinctive and less consistently retained than the head; its persistent parallel sixths and tenths also reduce melodic independence. Both main episodes rely on closely related neighbor figures and descending-fifth procedures, with the second constrained by its regular outer third-and-octave pattern. Local key areas receive only brief confirmation, and the stretto supplies a single concentrated test of the head's contrapuntal potential. Wide voicings at the C entry and the close reinforce register contrasts while thinning the middle of the texture.

Those limits support the description of a compact fugue or fughetta. The Bach comparison rests on thematic recurrence, functional bass motion, controlled suspensions, and a late increase in imitative density. The piece explores less of the sustained canonic writing, multiple countersubjects, and thematic transformations found across Bach's output. Its final cadence closes a concise exploration of one cell, with the return's changed temporal relation carrying most of the formal argument.

Sources and score reference

Pitch names use scientific pitch notation, with middle C designated C4. Half-beat references identify eighth-note positions in 4/4. Musical claims above refer to the final 26-bar source, including the two-entry stretto in mm. 19–20 and the arpeggiated close in mm. 25–26. The note evidence was checked against the LilyPond source and the corresponding exported MIDI; the MIDI cannot establish note spelling or the harmonic meaning of an interval.

- **Score:** GPT-6 Astra. 2026. *Fuga inversa*, for a single manual, in three voices. LilyPond source. The accompanying excerpt and evidence file records the source and MIDI hashes for this edition.
- **Gran, Jacob.** 2024. "A General Method for Composing a Canon Against a Cantus Firmus Using Sergei Taneev's Double-Shifting Counterpoint." *Music Theory Online* 30 (2), June. Article and examples. DOI: 10.30535/mto.30.2.4, especially §§3.7–3.8.
- **Hutchinson, Robert.** *Music Theory for the 21st-Century Classroom.* Website copyright 2017. §10.9, "Suspension"; §16.3, "The Cadential Six-Four Chord"; §29.3, "Voice Leading the Neapolitan Chord"; §30.8, "Fugue Analysis". The BWV 847 comparison refers to the score examples and commentary in §30.8.
- **Judd, Hannah.** 2022. "Virals, Memes, and the Lick's Circulation through Online Jazz Communities." *Twentieth-Century Music* 19 (3): 393–410. Published online 28 November 2022. Publisher abstract and article record.
- **Prout, Ebenezer.** 1891. *Fugue*, chapter 10, "Fughetta and Fugato," especially §§350–353. Public-domain text.
- **University of California, Santa Barbara.** "Fugue: Subject and Tonal Answer." Undated course handout hosted on Lee Rothfarb's faculty site. Two-page PDF.

External sources consulted 7 September 2026. The original score analysis, pitch tables, and evaluations are specific to *Fuga inversa*.

Appendix

Process and evaluation

This appendix records how the 26-bar *Fuga inversa* was composed, revised, and checked. The final score derives from candidate C, with a later stretto, a revised closing register, and explicit transfers of the middle voice between hands. The evidence consists of the LilyPond sources, compiled MIDI files, saved audit reports, candidate reviews, and performance scripts. The project records describe work undertaken on 7 September 2026.

Musical constraint and proposals

The user approved an exact inversion of the seven-note Lick, D–E–F–G–E–C–D. Reflection about its initial D gives D–C–B-natural–A–C–E–D. Its directed semitone intervals are −2, −1, −2, +3, +4, −2. The rhythm remains four eighth notes, a quarter note, and two eighth notes; the last note can be tied into the following bar. A diatonic adjustment from B-natural to B-flat would change the required interval sequence. This distinction affected the harmony throughout the composition.

The compositional working brief specified three monophonic voices on one organ manual, a literal tonic–dominant–tonic exposition, recurring counterpoint, episodes, and a credible return and cadence. The dominant answer therefore transposes the complete cell to A–G–F-sharp–E–G–B–A. Harmonic adjustments belong in the accompanying voices or continuation, while the recurring seven-note cell retains its pitches and rhythm under transposition.

At the user's request, three GPT-6 Astra composing agents worked at maximum reasoning effort. Each produced a separate candidate, compiled a score, and supplied notes for review. Their initial proposals were separate assignments, but the later revisions shared a brief, an auditor, coordinator feedback, and cross-reviews. This was a comparison of compositions produced under a common method.

A and C recommended B. Both valued B's recurring two-bar countermelody, high-register entry, and clear rise and return. B recommended C because its episodes move between voices and its middle section travels through G, C, and F. The coordinator retained C. C's own review identified its regular phrase lengths and frequent parallel thirds and sixths as limitations relative to B. That objection remains relevant to the comparison. These recommendations concern versions that precede the final stretto and expanded closing register.

Revisions that changed the score

An early A audit found parallel octaves in bar 6: soprano C-sharp5–D5 above bass C-sharp3–D3, from the second half of beat 2 to beat 3. The later A source revises those notes. The saved reports preserve both stages of this correction.

C's first saved audit covered only ten bars. Its adjacent-event scan found no perfect-interval flag, yet bar 10 contained two problems. On the quarter-note grid, alto C4 above bass F2 moved to E4 above A2, producing successive fifths. The intervening eighth-note sonority prompted a review of the underlying motion. At beat 4, G2–B3–D5 also exceeded the intended hand reach: either adjacent pair required more than an octave. The final version revises both the approach and the sonority, reaching B2–G4–D5 at that beat. The upper pair then fits comfortably within an octave.

The closing revision changes musical direction without adding a voice. The final soprano reads:

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d'4 cis'4 e'8 a'8 cis''8 e''8 | d''1
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The line rises through the dominant arpeggio to the final D5. Below it, F3 moves to E3 over A2 in bar 25, then to F-sharp3 over D3 in bar 26. The cadential six-four resolves before the arpeggio; the final chord contains exactly D3, F-sharp3, and D5. The wider register therefore comes from a single melodic line and an open three-note ending.

A later request for greater ambition led to a separate stretto study. The earlier C review correctly stated that its version contained no stretto. The new study tested a D entry in the soprano against a complete A answer beginning one eighth note later in the alto. Both heads retain the prescribed intervals and rhythm. They overlap for the remaining three and a half beats of bar 19, with the alto's final A4 falling on the first eighth of bar 20.

The study included unchanged bars 18 and 21 so that the approach and exit could be examined. It was engraved, compiled, and audited before the coordinator adopted its replacement for bars 19–20. The complete score was then compiled and audited again. This adds a ninth literal head within the existing 26 bars. The return now layers a tonic subject and dominant answer over an independent bass.

The alto's F-sharp4 on the second half of beat 2 in bar 19 belongs to the exact G4–F-sharp4–E4 descent. It is consonant above D3, then descends to E4 as the bass moves to C3. The D-major sonority accommodates the overlapping heads through this linear connection to C major.

What the audit tests

The auditor uses the actual LilyPond MIDI as its input. It groups notes by MIDI track and channel, checks monophony and crossings, and examines every voice pair at successive note events. It flags successive perfect fifths and octaves, distinguishes similar from contrary motion, and reports direct outer perfect intervals when the upper voice leaps by more than a whole tone. A second pass samples quarter-note positions. The two resolutions caught different problems in the drafts.

The final report identifies three monophonic voices and nine literal heads, with no flags from either perfect-interval pass and no crossings. Every simultaneous sonority permits an allocation between two hands within an octave. This last result concerns possible allocations; a separate check of the printed transfers is described below. The report's SHA-256 digest identifies the compiled symbolic MIDI to which the findings apply.

The subject matcher checks directed pitch intervals and relative attack times: 0, ½, 1, 1½, 2, 3, and 3½ quarter-note units. Source inspection supplies the written durations, ties, and enharmonic spelling that this MIDI-based match leaves unresolved.

Dissonance reports were therefore reviewed against the score. In bar 17, for example, the soprano A4 is prepared over F3, held while the bass moves to B-flat2, and resolved to G4 halfway through beat 2. The preparation and resolution explain the flagged seventh as a 7–6 suspension. Likewise, diminished sonorities need their resolutions examined. A stepwise melodic shape alone does not establish that a note is a permissible passing tone.

The auditor's fixtures include deliberate parallel fifths and octaves, a direct outer perfect interval, and examples of thirds and oblique motion that should pass. The tests passed for those specific program behaviors.

A later read-only review by Claude Fable 5.1 (`claude-fable-5-1`) examined published commit 7a156b0, including source, prose, PDF proofs, screenshots, and supplied test evidence. The reviewer did not listen to the recordings, operate the site, or execute the tests. Its findings were checked against the score before revision. That check also corrected two details in the review: the C-entry gap reaches a major thirteenth, and three of the four double-octave arrivals in bars 13–14 use similar motion. The review therefore supplied questions for investigation as well as proposed corrections.

Notation and performance

The abstract reach check asks whether one adjacent pair of the three sounding pitches can fit within an octave, leaving the other pitch to the other hand. The final score explicitly puts the alto in the left hand at bars 3–4, the end of bar 9, bars 11–12, and bars 25–26. At bar 9 beat 4, the low F-sharp3–G3 belongs with the left hand; at bar 11, C4 above bass C3 gives an octave. The closing F3–E3–F-sharp3 remains below the rising soprano.

A separate checker reads those printed staff assignments and transfers, matches all 382 notes to the MIDI, and checks each interval bounded by a note attack, note ending, or hand transfer. The maximum span is 12 semitones in each hand, including held notes. A deliberate omission of the alto's left-hand transfer at bar 11 makes the checker fail: the right hand would require C4–E5 at the downbeat and G3–E5 at beat 2½, spans of 16 and 21 semitones. The abstract allocation check still passes that fixture because its pitches are unchanged. This comparison tests whether the printed instructions provide the allocation that the notes require.

A MIDI fixture exposed a separate technical issue. LilyPond's voice channel setting still reused channel numbers across different staves. The audit could distinguish those notes by track and channel, but rendering required globally distinct voice channels. The preparation script remaps them and asserts that pitches and attack times survive. All organ voices then use the same registration.

The printed score and the MIDI export also serve different purposes. The printed score has two staves with hand transfers. A separate MIDI score keeps each contrapuntal voice on its own staff, preventing those printed transfers from splitting its identity. The symbolic MIDI remains the authority for interval analysis. Performance changes belong in derived files.

The organ realization uses Joseph Basquin's Jeux d'orgues 2.1 samples of the Stiehr–Mockers organ at Romanswiller. All three voices share the principal 8' + 4' registration. An 18-millisecond release gap supplies modest separation, while the tempo remains quarter note = 92 until the final bar slows to 60. The registration belongs to one manual; no pedal division is used. Instrument and samples.

The first piano realization used the Salamander Grand Piano SoundFont. The user found it too mechanical, despite its existing tempo, velocity, and articulation changes. That feedback prompted two separate revisions: a more pronounced interpretation and a different piano instrument. The final piano MIDI follows a smooth tempo curve, with the episodes approaching quarter note = 98–100, relaxation around the suspensions, and a broader cadential approach. The last whole note is held at quarter note = 54. The curve follows formal and harmonic events rather than random timing variation.

Note-on velocities range from 50 to 89. The entering head receives prominence even when it lies in the bass or alto; accompanying voices remain quieter. The descending head, local peaks, prepared dissonances, and resolutions have distinct attack weights. Releases vary with note length, melodic role, repetition, leap, and phrase boundary. Selected arrivals have small offsets, up to 17 milliseconds. Seven short harmonic spans use the damper pedal, with the remaining counterpoint kept clear of sustained pedal. The source and derived MIDI retain the same 382 pitches in the same order within each voice.

The revised MIDI drives GarageBand's installed Steinway Grand Piano. The complete performance uses one piano track, which gives the instrument a common pedal and acoustic identity. The native export disables the patch's reverb, echo, and compressor. The full mix and three separate voices were exported at 44.1 kHz and 24-bit PCM, with automatic normalization disabled. Their frame counts and embedded tempo markers agree. Native piano-roll observations preserve all pitches and place attacks within one native tick of the imported MIDI. Note lengths and velocities were checked in the imported MIDI; the native project supplied no round-trip check of those values. Separate voice renders use the same performance times and gain for analytical listening. The complete master remains the default recording; isolated voices are study aids.

The room treatment convolves a mono send with James Cadwallader's measured stereo response from Spokane Woman's Club, while retaining the original stereo piano as the direct signal. The OpenAIR metadata identifies an X/Y microphone at the rear of the hall, 24 metres from the stage source. The blend creates an artificial listening perspective from those measured reflections and the dry piano. OpenAIR dataset.

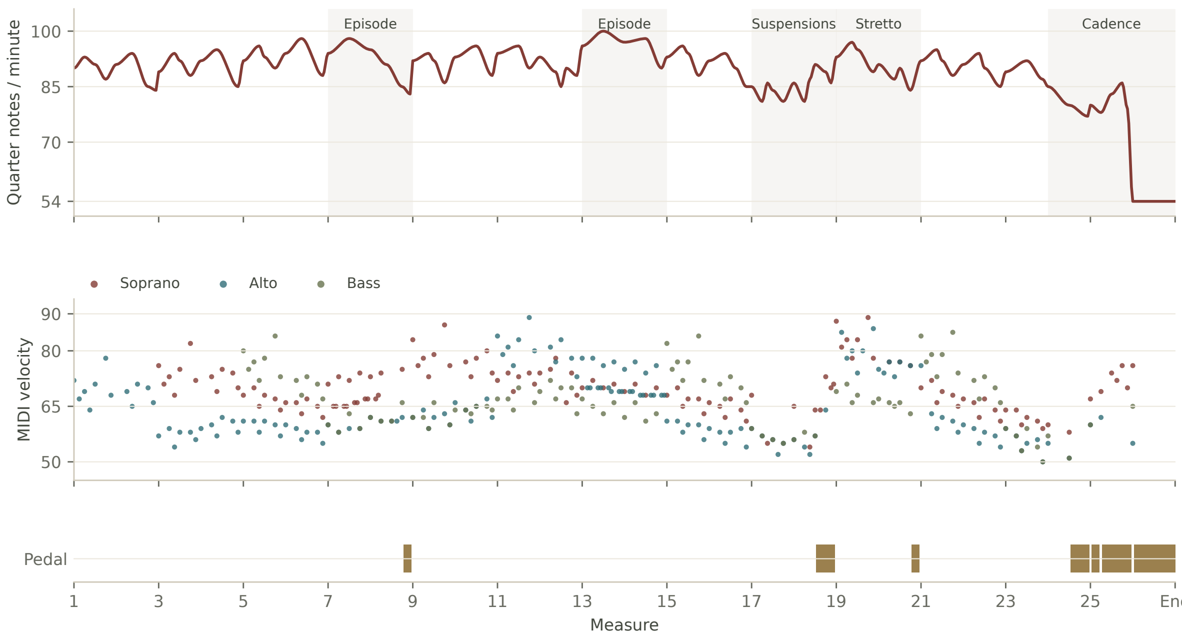

Figure A1. The final piano MIDI's tempo, note-on velocities, and pedal spans. Velocity is a MIDI control value, not a measurement of recorded loudness.

The processor removes the response's direct impulse, admits the measured reflections through a short fade, and filters only the room signal below 140 Hz and above 8 kHz. The retained response lasts 3.5 seconds. Its original broadband decay yields a 2.23-second T60 estimate from a fit over the −5 to −25 dB interval; this is an extrapolated acoustic measurement. The principal mix adds the room at −14 dB relative to the dry signal's RMS level, with 6 milliseconds of extra delay. A closer alternative uses −18 dB and 12 milliseconds. The alternatives are matched for integrated loudness, using static gain. No compressor or limiter is added to the piano master. The response is credited to James Cadwallader and OpenAIR under CC BY 4.0, with the processing changes identified in the edition's credits.

Digital edition

The musical examples are regenerated from the authoritative LilyPond source. A strict parser retains pitches, spellings, durations, rests, and ties; each compiled excerpt is compared with the corresponding full-score MIDI range. LilyPond 2.26's SVG output attaches source note identifiers to the engraving. The webpage uses those identifiers for analytical brackets, note explanations, and voice focus. It follows the selected recording through a time map that retains tempo changes within a beat. The organ and piano can therefore switch at the same score position.

The enhanced player lowers organ playback by 2.01 dB, matching the two masters' measured integrated loudness; downloads retain their original levels. Pause uses a 10-millisecond fade. Repeated excerpts taper for 5 milliseconds at each boundary, while whole-piece playback retains the recording's final decay.

Verovio was tested as a JavaScript notation renderer, including its semantic note identifiers. The final edition uses LilyPond SVG because the performing score was already authoritative and the paper needed short excerpts with matching printed figures. This choice preserves one engraving source. It gives up live musical reflow: phone readers scroll the fixed excerpts horizontally. The PDF uses the same annotated vector figures, typeset with XeLaTeX. LilyPond SVG output, Verovio semantic SVG.

Evidence and limits

The interval audit tests specified note relations; harmonic function and contrapuntal weight still require score analysis. Its quarter-note grid is one possible reduction. The agents' use of the same auditor leaves a shared source of possible error, despite their separate readings. The early reports cover unequal draft lengths, and the candidate recommendations concern earlier versions.

The printed-hand check establishes instantaneous reach, including sustained notes. Fingering, successive hand motions, articulation, and fluent legato remain untested by a live player. The MIDI channel fixture establishes a routing issue; it does not document an audible cutoff in the candidate recordings. These are separate questions from the score's formal and stylistic qualities.

The initial candidate recommendations came from score and MIDI inspection; A and C explicitly recorded no rendered-audio comparison. The production agents checked source data, render settings, waveforms, and loudness, but supplied no auditory evaluation of the final recordings. The user's listening feedback changed the production plan. The room treatment combines a measured response with a separate instrument recording, without documenting a concert or a particular seat. The checks cover specified pitches, intervals, timing, instantaneous hand spans, and signal levels; assessing musical effect and performance demands still requires listening and a player's trial.

Primary artifacts

Paths are relative to the edition's reproducibility directory. The first-draft MIDI files were not retained; their saved reports preserve historical findings without making those drafts rebuildable.

Artifact	Evidence
`fugue.ly`; `fugue.midi`; `reports/final-audit.json`	Final notation, compiled notes, and interval findings tied to a MIDI digest
`candidate-a/NOTES.md`; `candidate-b/candidate-notes.json`; `candidate-c/REVIEW.md`	Candidate forms, limitations, and differing recommendations
`history/index.json`; `history/candidate-*-*-audit.json`	Saved draft findings, with retained-file limits recorded separately
`paper-process/stretto-study.ly`; `paper-process/stretto-study-audit.json`	The adopted overlap with its approach and exit
`audit_midi.py`; `fixtures/channel-fixture.ly`	Check definitions and the channel example
`check_hands.py`; `reports/printed-hands.json`; `fixtures/missing-alto-transfer.ly`	Printed hand allocation and a failing transfer example
`piano-production/shape_performance.py`; `piano-production/performance-checks.json`	Tempo, articulation, dynamics, pedal and the source-to-performance comparison
`piano-production/garageband-import-checks.json`	Native pitch and onset checks, with round-trip limits stated
`hall-production/hall.py`; `hall-production/public-provenance.json`	Measured room processing, source hashes, native formats and delivery measurements

The complete file manifest accompanies this edition. The musical claims above should be read together with the checks' stated limits. The score and recordings and source repository are available online.